

## Design, Simulation & Analysis of Forces exerted on a New Modeled Unmanned Aerial Vehicle.

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### Abstract

*In this paper we will discuss about simulation & analysis of a newly designed model of UAV which has been specially made for Vertical Take-off & Vertical Landing (VTOL) and as well as a mode which converts it to a high speed flight mode UAV which has been never been designed yet with a better feasibility factor. The old forms of Quad-rotors have only the VTOL advantage but fail to escape from sight when the enemy falls into the view, thus lacking the stealth mode. And apart from mechanical side, it has features like GPS tracking, Live video streaming, High point stability, Stealth modes and more features of the UAV, it had fail safe modes like, if the drone is with one damaged motor, then the back or front part actuators can individually bring the drone back to base station acting as the backup and main drive.*

**Keywords:** UAV, Drones, Flight mode, VTOL Mode, Booster Motors

### 1. Introduction

The unmanned aerial vehicle (UAV) is a small plane which flies in an autonomous way with only one virtual pilot on board. That does not mean that the plane is completely out of control once it leaves the track. The presence of a human being in the loop of control is essential. But the operator is on a station on the ground, either on ground, or at sea or in a control center. He can intervene at any time, examine the situation and the rules of engagement, and stop the mission if necessary. The concept of an UAV resonates in several research laboratories of any developed country. The first instigators of this type of the projects were the soldiers. Indeed, the armies of many countries prefer to rather send UAV's for the recognition in the high-risk zones than man. This type of flying machine can take much various missions, extending from the control of the scientific experiments to the intelligence collecting the monitoring during the day or the night. The numerical simulation in the field of

aerodynamics is relatively recent research tools. The specialists raise the question of knowing if the field of aerodynamics is purely theoretical or numerical, they estimate however that it is theoretical, because of the many numerical tests necessary for the stage to insufficiencies of the knowledge of the calculation methods. Since a score of years, one can say that numerical aerodynamics lost much of its empirical nature due to the convergent efforts of mathematics to solve the nonlinear problems involved in the calculation methods in aerodynamics and also thanks to the invention of the very fast machines of calculation and the development of data processing.

In this present work, a graphical simulation of a force around an UAV and its subsystems is presented. The first phase of this work consists in using plans of construction of this UAV to create a solid model which specifies its external geometry in Solid Works. This solid model is then exported in a programme like Autodesk Inventor of the grid generation in order to return its geometry adapted to aerodynamic calculations and tested in various flying conditions. A computer code of computational fluid dynamics (CFD) is used to obtain estimates of the coefficients of lift and drag to 10 meters per second. A study of the independence of the grid is then carried out to determine the precision of the grid of calculation used for analysis CFD and other airflow stress simulations.

### 2. Advancements

In normal UAV's like the Reaper and Panther it contains the universal flight mode but lacks fail safe modes of landing at high terrain places during emergency thus weakening their part. Now combining both the features brings to an ultimate UAV which proves a lot during the war field. But it's not that simple to design such a model as it need a lot of mechanical factors which falls good during aerodynamic conditions as a small force point at the non-pressure points will lead to damage and crash landing. There have been jet fighters like the Jet Harrier with VTOL and Flight modes but

they are for attack. And there have been a lot of designs for Surveillance with both these features but no one has yet brought down to a feasible and marketable application. We have worked on the mechanical design now which stands to be feasible as it being tested on the aerodynamic use as the mechanical design has been tested well. The aerodynamic and mechanical simulation are described here which will prove to its feasibility. More from the mechanical side, it can stop very swiftly during flight modes in the case of emergency. It does not have to take a gliding turn when needed, it just needs to stop at the point and turn at the point itself thus saving the flight time. In brief, in this abstract we'll discuss about some of the mechanical design, and its simulation and analysis. The photos are provided for a well virtual creation and analysis.

### 3. Description of the UAV

As you can see in Fig. 1, it's in the normal extended flight mode now. In the design, there are two movable wings at the end of which are connected the brushless motor and the tri blade. These wings are connected to the high torque servo motors which are installed in the hull of the drone. These wings stay vertical during the take-off or landing mode. Adjacent the point of brushless motor is there is another servo motor whose shaft is connected to the secondary wing which unfolds during the flight mode. These secondary wings help for gliding in the air thus saving the power of the battery. Now to the back of the drone, we have the booster brushless motor which gives the speed to the UAV. They are also movable to a resolution of 90 degrees. During VTOL mode Fig. 2, it's also in the vertical position, and this feature of having it vertical brings to the fact that it won't restrict the direction of take-off. This will cut down the wind and pause and instability coming in its way.



Figure 1. Flight Mode

Now during the flight mode the main wings rotate 90 degrees to horizontal position and the secondary wings also retract or extend which brings it to the flight mode. The

booster motors also rotate to horizontal positions and applying the necessary speed to the glider.



Figure 2. Vertical Lift off and Landing Mode

This UAV as shown in Fig. 3 is 2 meters long and provided with an elevator wing of the scale of 40 m, the profile of this wing is of type NACA 2412 with a cord of 0.236 m, stalled on the fuselage of 3° with a dihedral of 2°. The elevator group of the shape of T reversed is profiled with Naca 0012 and attached to the wings by two beams which give also the name of an UAV to tristar. These UAV's aspire of a remarkable stability to the no desirable movements of roll. This UAV is designed to have a booster motors which will be embarked behind the fuselage.

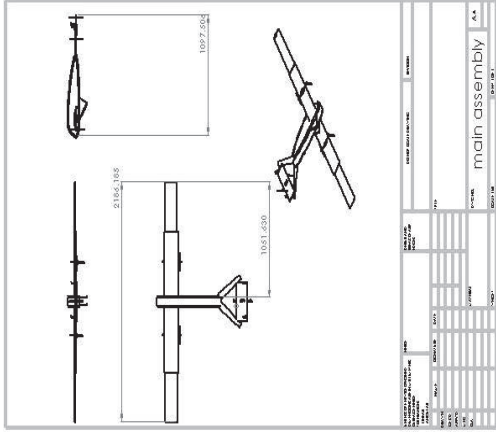


Figure 3. 2D CAD Model

#### 4.1. Airfoil Wind Tunnel Test

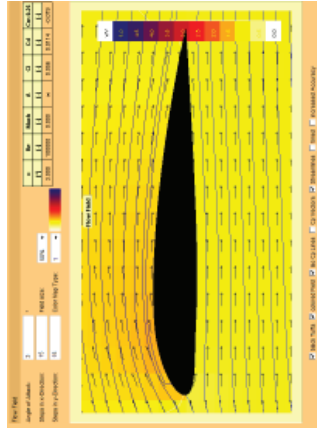


Figure 4 Velocity Ratio Analysis

Several compilations were carried out for various angles of incidence to determine the lift and drag. Each case was carried out at the same speed: 10 m/s. The angle of attack extended from 3 to 18 degrees. It is a parabolic type tendency similar to those found on the planes in general. In particular, one notes the increase in the drag with the increase in the angle of incidence. This curve is presented with drag of the isolated wing which equips the UAV in question and which was obtained by wind tunnel test. The two diagrams follow the same tendency separately which is normal because variation of the drag between the two curves corresponds to the drag of the remainder of the UAV which varies very slightly with the incidence. The current NACA airfoil proves to be having a low turbulence and a better lift factor with the angle of attack set at 3degrees. The velocity ratio is balanced properly. And the pressure generation around the airfoil is distributed symmetrically converging into a stable airfoil design.

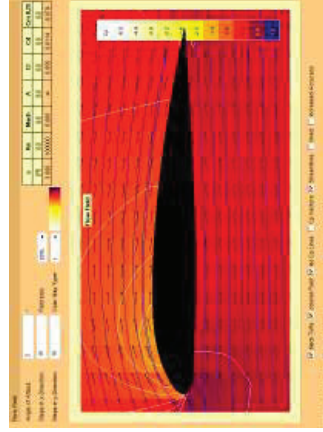


Figure 5 Pressure Co-efficient Analysis

#### 4.2. Eppler & NACA Analysis

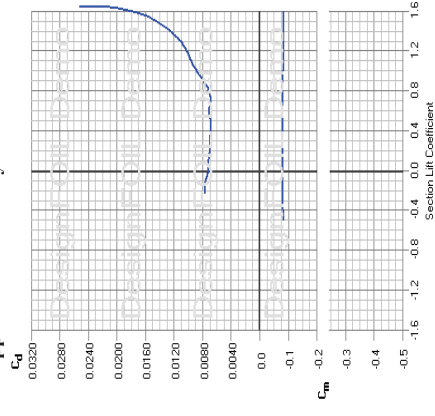


Figure 6 NACA Analysis of Cd vs Cl

An agreement of the two curves is noted in the linear zone which is explained by the fact that it is the wing which contributes in a great proportion to the total lift produced on the plane. The other parts of the plane, in particular the stabilizer contribute slightly towards weak and moderate incidence. However the lift of the isolated wing exceeds that of the plane for the bigger angles of incidence. This is dues with the fact that the Reynolds number of the tests of the isolated wing is much higher than that of the plane tested numerically which leads to higher maximum  $C_m$  for the isolated wing. Numerical calculations show well the phenomenon of unhooking of the UAV which is located at approximately 3 degrees. It should be noted that this incidence which appears is weak only seemingly because it should be mentioned that the reference of measurement of the angle of attack is the base of the fuselage and that the wing is fixed with 4 degrees, flow return is noted on almost the totality of the wing, in particular, it is much attenuated in the central part.

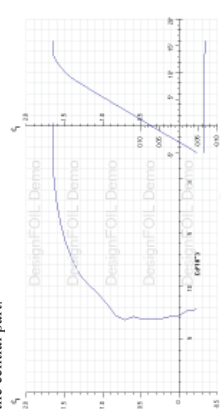


Figure 7 Eppler Analysis of Cd vs Cl

This effect is expected because the wing is not twisted and the great deflections of the air are perceptible in this area. The separation of the flow recorded on the nose of the fuselage still persists with this angle of incidence, flow recorded on the level of the wing root with the fuselage with 15° of incidence.

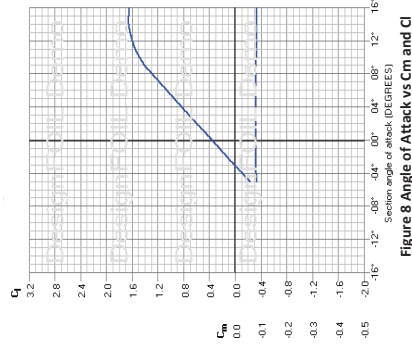


Figure 8 Angle of Attack vs Cm and Cl

#### 4.3. CFD Analysis of Fuselage & Airfoil

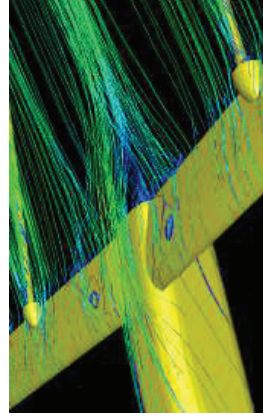


Figure 9 Air flow around Fuselage & Airfoil

It is a representation of the streamlines of the flow inside the internal zone of the boundary layer around the UAV which is completely pressure building. It is a very useful diagram for the draftsman of the aerodynamic forms because it reveals the design defects characterized by the appearance of the zones of flow separated color blue as shown in the Fig.9 on the wing and on the level from the bases from faster on from the wings. Fig. 9 shows the two marginal swirls which release from salmons of the two wings of the UAV to the passage of the flow for the incidence 15°, the size of the marginal swirl for such an incidence is considered to be

considerable. Moreover, one notes the complexity of the flow around the plane for an incidence judged like the beginning of turbulence.

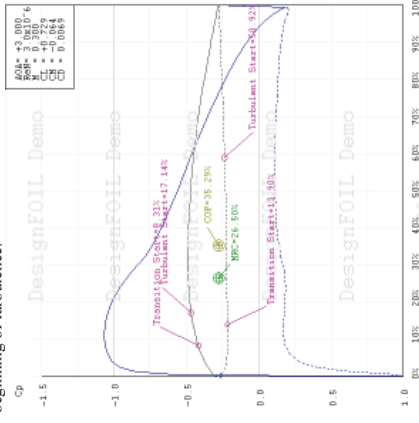


Figure 10 Turbulent Analysis

The trailing edge leads to a better flow with a slight generation of turbulence at the top camber line which is obvious as there generation of pressure built at that position.

#### 4.4. UAV Curvature & Smoothness Analysis

In the deterministic optimization study, wing airfoil shape, wing taper ratio, wing geometric twist angle, and wing quarter chord sweep angle are considered as design variables, as well as the lay-up scheme of the composite bendable wing. The airfoil at the root of the wing is completely defined by four variables, the camber towards the leading edge, the chord-wise position of this control point, and similar information for the camber towards the trailing edge. These variables are defined to be the chord normalized values.



Figure 11 VTOL Mode Curvature

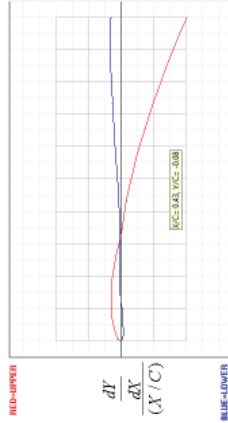


Figure 12 First Derivative Curve

A quadratic curve from the leading edge to the maximum camber point, which takes in the first derivative, a polynomial in the middle and another quadratic curve from reflex point to the trailing edge, can completely and uniquely define root airfoil shape and its smoothness. Curvature continuity is ensured between the color combinations. For a constant camber wing design the tip airfoil geometry is the similar as the root airfoil, while for a variable camber wing design the tip airfoil is defined independent of the root airfoil.



Figure 13 Flight Mode Curvature

All the metrics are interpolated between the root and the tip to obtain a kink free wing shape, as can be graphically seen above.

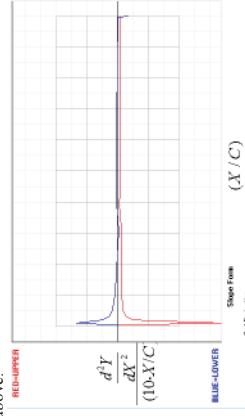


Figure 14 Second Derivative Curve

Due to manufacturing constraints (plain weave bi-directional nature of material used in the manufacturing) and to ensure symmetric deformation of the two wing halves under aerodynamic loading, only smoothened layup orientations are allowed.

#### 4.5. Wing Stress & Loading Analysis

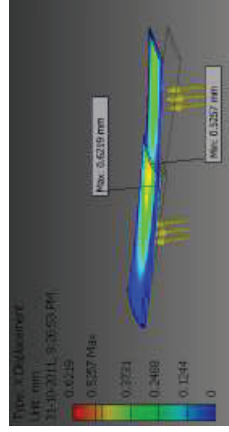


Figure 15 Wing Loading Effect

While both aerodynamic and structural metrics are included in the wing design, the analysis are not explicitly coupled in an aeroelastic sensor, stable deflections of the wing due to air loads are assumed to be small, since large ones will lead to buckling failure. Aerodynamic performance of the wing is studied using the fixed wing stress analysis. Aerodynamic analysis computes the pressure distribution over the wing, as well as the concomitant lift, lift-to-drag ratio (L/D) and the static stability margin. An arc length method based nonlinear deviation tool is used to evaluate the structural performance of the wing and to determine maximum flying velocity that the wing can withstand without buckling or failing (laminar failure with factor of safety = 1.5) under aggressive flight loads.

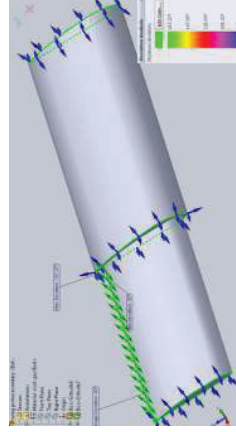


Figure 16 Deviation Analysis

Experimental validation of the numerical procedure used for determining maximum flying velocity has been carried out and can be found elsewhere. An analytical approach based on classical laminate plate theory and strain superposition (initial chord-wise strains due to root airfoil flattening and additional span-wise strains due to incremental rolling) is

used to study the stresses developed in the composite wing during the rolling process for storage with a safety factor of 1.5 is used to check first ply failure of the composite wing to determine minimum safe storage diameter.

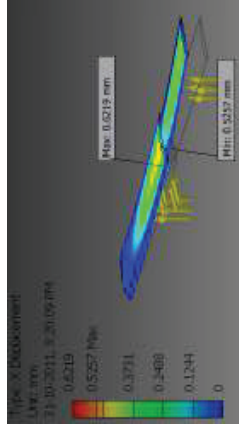


Figure 17 Thrust & Loading Effect

Taking advantage of problem symmetry, we model only one wing half, with the appropriate symmetric boundary conditions at the root airfoil. And appropriate results have been found as the stress analysis proved correct and the wing was able to sustain the pressure at the trailing edge as we rarely see reddened part on the airfoil. Reaching to a corrected figured value achievable to 10m/s or 20m/s.

#### 5. Conclusion



The present study was undertaken to observe the aerodynamic performances of an UAV by using a CFD computer code. This concept of design of flying machines is judged like an alternative to the wind tunnel tests for a possible reduction of the design cost. This work brought moreover, certain important aerodynamic corrections around this UAV such as the aerodynamic stability of the nose of the fuselage and the setting of the bases of fastener of the beams on the level of the under-surface of the wings with the proper stress analysis done.

Limitations relate to the power usage of the UAV during vertical take off mode which might drain off huge currents but would be compensated by the flight mode. New methods of endurance increase would be looked on.

Several change and simplification were made in the modeling process, including the use of electric brushless motors instead of reciprocating, the constant mass, counter-rotating propeller to eliminate rotating-mass effects, the removal of aileron control, etc. These changes were made in a way that the flight characteristics would not change much.

The simulations were conducted several times performing a take-off, hover, cruise and landing maneuver. Softwares like Solidworks, Autodesk Inventor, Design Foam and many others have contributed a lot for the analysis and simulations done in this paper. It's been a successful path along.

The future scope and the work that needs to be carried out is the hardware implementation and testing of the retractable wing, coming onto the mechanical point of view. Airfoil has to be tested well with the attachments of the fuselage. Control system has to be designed to meet this new design requirements stability.

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